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Effects of reading proficiency on embedded stem priming in primary school children



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ABSTRACT

Prior evidence from masked morphological priming has revealed conflicting findings regarding the acquisition of morpho-orthographic segmentation mechanisms in developing readers. Here, we examined changes in masked morphological priming across grade within a large sample of French primary school children ($n = 191$, Grades 2–5) and how these effects are modulated by individual differences in reading proficiency, spelling proficiency, and morphological awareness. Target words were preceded by either (a) a suffixed word prime (e.g., *tristesse-TRISTE*), (b) a suffixed nonword prime (e.g., *tristerie-TRISTE*), (c) a non-suffixed nonword prime (e.g., *tristald-TRISTE*), or (d) an unrelated prime (e.g., *direction-TRISTE*) using very short prime durations (50 ms). Moreover, a frequency manipulation was included for suffixes and non-suffixes. The results revealed robust suffixed word priming across all children independent of grade and proficiency. On the other hand, priming in the suffixed and non-suffixed nonword conditions was modulated by reading proficiency, with high-proficiency children showing facilitation and low-proficiency children showing inhibition. The effects of suffix and non-suffix frequency were modulated by grade, with decreasing effects as grade increased. None of the observed priming effects were modulated by grade, spelling proficiency, or morphological awareness. The results suggest that reading proficiency is an important predictor for embedded stem

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activation mechanisms in primary school children, which we discuss in the context of recent theories of morphological processing.

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Introduction

Over the past decades, much research has been dedicated to understanding how morphemes are processed in the skilled reading system. Most of this research has used the masked primed lexical decision task in combination with behavioral and sometimes electrophysiological measures. This research has generated important insights into the mechanisms and time course of morphological processing in adults (e.g., Beyersmann, Iakimova, Ziegler, & Colé, 2014; Diependaele, Sandra, & Grainger, 2009; Morris, Frank, Grainger, & Holcomb, 2007; for a review, see Rastle & Davis, 2008). However, only recently have scientists begun to explore the development of such processes in children (e.g., Beyersmann, Castles, & Coltheart, 2012; Casalis, Dusauroir, Colé, & Ducrot, 2009; Quémart, Casalis, & Colé, 2011). We first outline the key milestones from morphological processing research in adults, followed by a summary of the related evidence from reading development.

Morphemes are the smallest meaningful entities of which words are composed. Based on a growing body of psycholinguistic evidence in adults, it is now known that the activation of morphemes is a rapid, highly automatized process that does not necessarily require the involvement of semantics. This now broadly accepted view primarily originates from the results reported in numerous masked morphological priming studies, across different languages, showing that not only words with a true morphological structure (e.g., *farm-er*) but also words with a pseudo-morphological structure (e.g., *corn-er*) produce priming of the stem target (e.g., *farm*, *corn*) even if presented with very short (40–50 ms) prime durations (for a review, see Rastle & Davis, 2008). Because the meaning of a pseudo-affixed word like *corner* cannot be derived from the meaning of its constituents *corn* and *er*, it has been argued that the activation of the embedded stem *corn* must be due to a semantically blind “morpho-orthographic” decomposition process.

In addition to the evidence for morpho-orthographic processing, masked morphological priming studies in adults have further revealed that truly suffixed words tend to produce slightly stronger stem–target priming (e.g., *farmer*–*FARM*) than pseudo-suffixed words (e.g., *corner*–*CORN*), particularly when the prime is partially or fully visible (e.g., Diependaele et al., 2009; Feldman, O'Connor, & Moscoso del Prado Martin, 2009). These findings suggest that the early morpho-orthographic processing stages are followed by a later “morpho-semantic” processing stage by which truly affixed words (but not pseudo-affixed words) are segmented into semantically defined morphemic units.

Critically, the data discussed above have recently been complemented by masked priming studies using morphologically structured nonword primes. Longtin and Meunier (2005) were the first to report priming with suffixed nonwords (e.g., *rapidifier*–*RAPIDE*) but not with non-suffixed nonwords (e.g., *rapiduit*–*RAPIDE*). However, recent morphological priming studies have found results that partially deviate from the Longtin and Meunier pattern (Beyersmann, Casalis, Ziegler, & Grainger, 2014; Morris, Porter, Grainger, & Holcomb, 2011). Beyersmann, Casalis, and colleagues (2014) examined lexical decision times to target words preceded by one of four different types of primes: (a) a suffixed word prime (e.g., *tristesse*–*TRISTE*), (b) a suffixed nonword prime (e.g., *tristerie*–*TRISTE*), (c) a non-suffixed nonword prime (e.g., *tristald*–*TRISTE*), or (d) an unrelated prime (e.g., *direction*–*TRISTE*). The results revealed robust and equal priming in the two suffixed conditions across all participants, suggesting that not only suffixed real words but also suffixed nonwords are rapidly decomposed into morpho-orthographic subunits at early stages during visual word recognition. Thus, this demonstrates that morpho-orthographic decomposition mechanisms apply even in a nonword context where the semantic stem–affix relationship is entirely non-interpretable.

Most important, however, [Beyersmann, Casalis, and colleague's \(2014\)](#) findings revealed that priming in the non-suffixed nonword condition was modulated by individual language proficiency. Whereas high-proficiency adults showed as much priming in the non-suffixed condition as in the other two conditions, low-proficiency adults did not show any evidence for non-suffixed priming. The priming effect observed in the non-suffixed nonword condition is of particular relevance because it suggests that high-proficiency individuals rapidly activate embedded stems at early stages during visual word recognition even in the absence of an affix. Hence, these nonword priming data suggest that affix stripping in adult readers is complemented, and possibly even substituted, by embedded stem activation mechanisms.

Prior evidence from lexical decision indicates that children already use morphological structure from as early as 7 years of age ([Lázaro, Camacho, & Burani, 2013](#); [Perdijk, Schreuder, Baayen, & Verhoeven, 2012](#)). However, recent findings from masked morphological priming suggest that morpho-orthographic decomposition is quite a late-occurring milestone in reading acquisition ([Beyersmann et al., 2012](#)). One hypothesis that has been put forward is that children first learn to link the meanings of related whole words (e.g., a *painter* is someone who *paints*, a *teacher* is someone who *teaches*), which eventually results in a form-based morphological segmentation mechanism that is applied to every letter string with the mere appearance of morphological complexity (e.g., *moth-er*, *numb-er*, *flow-er*). Critically, previous studies using the masked priming paradigm to examine the role of morphological knowledge during reading development have revealed conflicting findings. [Quémart and colleagues \(2011\)](#) tested French third, fifth, and seventh graders and found evidence for true morphological priming (e.g., *farmer*–*FARM*) and pseudo-morphological priming (e.g., *corner*–*CORN*) across all age groups. Thus, the findings replicate the pattern standardly observed in adults, showing that French children rapidly decompose complex words into semantically blind morpho-orthographic sub-units as early as Grade 3. However, [Beyersmann and colleagues \(2012\)](#) observed a different pattern in two groups of English-speaking third- and fifth graders. Masked primes were presented for 50 ms, and priming effects were found only for prime–target pairs sharing a true morphological relationship (e.g., *farmer*–*FARM*), but not in the pseudo-morphological condition (e.g., *corner*–*CORN*). Thus, Beyersmann and colleagues' results demonstrate that English-speaking third- and fifth graders produce priming only when prime and target are semantically related, suggesting that they have not yet acquired abstract morphemic representation that is thought to be essential for the automatic segmentation into morpho-orthographic units.

One possible reason for these divergent findings is the fact, first pointed out by [Baayen, Milin, Filipovic Durdevic, Hendrix, and Marelli \(2011\)](#), that prime words previously tested in the opaque/pseudo-morphological priming condition vary in the extent to which the suffix conveys its regular meaning. For example, as [Baayen and colleagues \(2011\)](#) suggested, the etymological origin of *archer* (“someone who wields a bow”) is Latin *arcus* (*bow*, *arc*), which is similar in structure to a denominal formation such as *trucker* (“someone who drives a truck”). On the other hand, a word like *corner* is clearly pseudo-suffixed because the meaning of *corner* cannot be derived from the meaning of *corn* and *er*. In the current study, we avoided this problem by using complex nonword primes to test for morpho-orthographic segmentation in children, as in the prior research with adults summarized above ([Beyersmann, Casalis, et al. \(2014\)](#)). It is also possible that morphological parsing mechanisms in French and English are acquired at different speeds. French is morphologically more productive than English, and French children have been found to outperform English children in morphological knowledge (e.g., [Duncan, Casalis, & Colé, 2009](#)). Therefore, it is possible that French children begin to automatize morphological processing at an earlier age than English children, which would explain the absence of morpho-orthographic priming in English-speaking readers. A third fundamental confound within the assessment of morphological processing mechanisms in developing readers may be that children within the same age/grade group vary widely in their individual proficiencies in reading, spelling, and morphology. For instance, some children during the last years of primary school have reading proficiency levels that are comparable to those of children in the lower grades and vice versa. One solution is to exclude children with reading problems who fall outside the expected proficiency range per grade level. However, this approach not only results in the loss of valuable statistical power but also narrows the analyses down to grade-level comparisons rather than using more fine-tuned continuous proficiency measures irrespective of grade. Therefore, to more clearly capture the

relationship between the acquisition of morpho-orthographic parsing mechanisms and the development of reading skills, it is essential to explore individual differences in reading proficiency continuously rather than analyze children per grade. Similarly, individual differences in spelling proficiency and morphological awareness may present confounds in previous developmental priming studies. However, masked priming evidence in children is sparse and based on rather small participant samples from selected age groups (e.g., [Beyersmann et al., 2012](#); [Quémart et al., 2011](#)). Thus, the role of individual proficiency differences during the acquisition of morphological processing mechanisms is still not clear.

The goal of our current study was to examine the early processing of embedded stems and affixes within the developing reading system. Here, we tested a large sample of French primary school children ($N = 191$ from Grades 2–5) with varying levels of reading proficiency, spelling proficiency, and morphological awareness to explore whether or not these individual differences modulate morphological priming. We replicated the design previously used in adults ([Beyersmann, Casalis, et al. \(2014\)](#)), comparing target words preceded by either (a) a suffixed word prime (e.g., *tristesse*–*TRISTE*), (b) a suffixed nonword prime (e.g., *tristerie*–*TRISTE*), (c) a non-suffixed nonword prime (e.g., *tristald*–*TRISTE*), or (d) an unrelated prime (e.g., *direction*–*TRISTE*), using very short prime durations (50 ms). Critically, in contrast to previous developmental work ([Beyersmann et al., 2012](#); [Quémart et al., 2011](#)), we used the exact same targets in all four prime conditions to rule out that differences across conditions might be due to uncontrolled differences between target words.

If affixes form the primary recognition units in the developing reading system, we would expect significant priming in the suffixed word and suffixed nonword conditions but not in the non-suffixed condition. However, if children also draw on an embedded stem activation mechanism during the early stages of visual word recognition, we would expect significant priming in all three priming conditions (including the non-suffixed condition). If morpho-orthographic segmentation mechanisms are acquired over the course of primary school, we would expect to see an increase of suffixed word and suffixed nonword priming across grades. If embedded stem activation mechanisms are acquired over the course of primary school, we would expect to see an increase in all three priming conditions across grades. Moreover, if morphological priming is modulated by children's performance in reading, spelling, and morphological awareness tasks, we would expect that the magnitudes of suffixed nonword and non-suffixed nonword priming should vary depending on individual differences in these scores. In addition, given that previous studies in developing readers have unambiguously reported evidence for true morphological priming effects (e.g., [Beyersmann et al., 2012](#); [Quémart et al., 2011](#)), we expected that priming in the suffixed word condition should be robust across all children independent of proficiency.

Finally, we adapted a further feature of [Beyersmann, Casalis, and colleagues' \(2014\)](#) materials, where suffixes and non-suffixes were divided into high- and a low-frequency conditions. If it is true that affixes form morpho-orthographic units in the developing reading system (e.g., [Quémart et al., 2011](#)), we would expect more priming for primes with high-frequency suffixes than low-frequency suffixes but no difference between high- and low-frequency non-suffixes. However, if suffixes and non-suffixes have a comparable status during the first years of reading development, we would predict that high ending frequency should boost the activation of embedded target words independent of the morphemic status of the ending string.

Method

Participants

A total of 191 children in Grades 2 to 5 were recruited from a primary school in Lille and Arras in northern France (46 second graders, 48 third graders, 49 fourth graders, and 48 fifth graders). The mean participant age per grade was 7.7 years in Grade 2, 8.7 years in Grade 3, 9.6 years in Grade 4, and 10.6 years in Grade 5. All children were French native speakers with normal or corrected-to-normal vision. Parental consent was obtained prior to the testing. Children with a history of dyslexia and/or language impairment were excluded. No child was affected by hearing loss.

Materials

The materials were identical to those used in [Beyersmann, Casalis, and colleagues' \(2014\)](#). We first confirmed that the 50 word targets and suffixed word primes, originally selected from the adult corpus Lexique ([New, Pallier, Brysbaert, & Ferrand, 2004](#)), were also present in a lexical database for first graders (MANULEX; [Lété, Sprenger-Charolles, & Colé, 2004](#)). Each target was preceded by a suffixed word prime (e.g., *tristesse*–*TRISTE* [sadness–SAD]), a suffixed nonword prime (e.g., *tristerie*–*TRISTE* [sation–SAD]), a non-suffixed nonword prime (e.g., *tristald*–*TRISTE* [sadald–SAD]), or an unrelated prime (e.g., *direction*–*TRISTE* [direction–SAD]). Suffixed word primes were real words comprising stem (e.g., *triste*) and suffix (e.g., *esse*). Suffixed nonword primes were created using the same stem but a different suffix of comparable frequency (e.g., *rie*), such that the whole letter string was not a word. Non-suffixed nonword primes were created by combining the stem with a non-morphemic ending (e.g., *ald*). Unrelated primes were suffixed words with all letters different from the target. All non-words were orthographically legal and pronounceable. The four prime conditions were matched on length (for a full list of materials, see [Beyersmann, Casalis, et al. \(2014\)](#), [Appendix 1](#)).

Trials were divided into two subsets: one where primes comprised high-frequency suffixes (*-ette*, *-ier*, *-eur*, *-ion*, and *-age*) and one where primes comprised low-frequency suffixes (*-esse*, *-oir*, *-ade*, *-ien*, and *-rie*). Non-suffixed nonwords were created using five high-frequency non-morphemic endings (*-nie*, *-ire*, *-ide*, *-ert*, and *-use*) and five low-frequency non-morphemic endings (*-uor*, *-ald*, *-abe*, *-uto*, and *-bli*). String frequency was calculated based on the number of words in the MANULEX database ([Lété et al., 2004](#)) that contained the given letter sequence in final position. The frequency of non-suffixes ranged between 2 and 347, and the frequency of suffixes ranged between 74 and 581. Suffixed words and nonwords used the same suffixes but with different stems. High- and low-frequency conditions were matched on length, frequency, suffix length, and non-morphemic ending length. For the purpose of the lexical decision task, 50 nonword targets (identical to the Set 2 nonwords in [Beyersmann, Casalis, et al. \(2014\)](#)) were included and matched to the word targets on length.

Procedure

The procedure was identical to the one used in adults ([Beyersmann, Casalis, et al. \(2014\)](#)). Stimuli were presented in the center of a computer screen using DMDX ([Forster & Forster, 2003](#)). Each trial consisted of a 500-ms forward mask of hash keys, then a 50-ms prime in lowercase, and then the uppercase target. The target remained present until the response or until 3 s had elapsed. Children were instructed to respond as quickly and accurately as possible. The experiment took between 3 and 6 min per participant.

Measures of individual differences

On completion of the masked primed lexical decision task, each child was assessed with a spelling proficiency test, a reading proficiency test, and a morphological awareness test. The morphological awareness test was used to measure children's awareness of morphological structures during spoken language processing. The reading and spelling tests were used as two measures of children's proficiency in processing printed words.

Spelling proficiency

The "Corbeau" subtest was used to assess spelling proficiency in third to fifth graders ([Chevrie-Muller, Maillart, Simon, & Fournier, 2010](#)). For each child, three different scores were obtained: (a) number of responses with correct phonetic form (e.g., *oizo* instead of *oiseau*), (b) number of responses related to existing lexical entries (e.g., *oiseaux* instead of *oiseau*, *rendu* instead of *rendus*), and (c) number of correct responses. Spelling proficiency in second graders was examined based on a different dictation test ("Dictée de phrases") adapted for children with little spelling experience ([Chevrie-Muller et al., 2010](#)).

Reading proficiency

The “Alouette” was used to assess children’s reading proficiency (Lefavrais, 1974, 2005). Children were asked to read aloud a text of 265 words for which both reading accuracy (*l’indice de précision*) and speed (*l’indice de vitesse*; how many words are read within 3 min) were measured and evaluated based on the revised Alouette 2005 scoring system.

Morphological awareness

Morphological awareness was assessed with a test developed by Casalis, Colé, and Sopo (2004) consisting of three different spoken language tasks: two sentence completion tasks and an oddity task. In the sentence completion tasks, children were asked to produce a real affixed word (e.g., “Someone who works on a farm is a ...”) or a pseudo-affixed word (e.g., “Someone who works in a birk is a ...”). In the oddity task, children were presented with four orthographically related words, among which only three shared the same morphemic stem (e.g., *turn*, *turning*, *turner*, *turnip*). The score was the number of correct responses in each task.

Results

Lexical decisions to word targets were analyzed as follows. Incorrect responses were removed from the reaction time (RT) analysis (11.4% of all data). Outliers below 200 ms were removed, leading to the removal of 0.03% of the data. 1/RTs were calculated for each participant to correct for RT distribution skews and used throughout the analyses. Mean untransformed RTs and error rates were calculated for each reading age level (based on the individual Alouette reading proficiency scores; Lefavrais, 2005) and are presented in Table 1.

We used linear mixed-effect modeling to perform the main analyses (Baayen, 2008; Baayen, Davidson, & Bates, 2008). Fixed and random effects were included only if they significantly improved the model’s fit in a backward stepwise model selection procedure (i.e., factors were progressively dropped from the model based on statistical significance). Models were selected using chi-square log-likelihood ratio tests with regular maximum likelihood parameter estimation. Trial order was included to control for longitudinal task effects such as fatigue and habituation. To assess whether the obtained effects were modulated by individual differences of reading proficiency, spelling proficiency, or morphological awareness, the scores from the spelling dictation, reading aloud, and morphological awareness tests were standardized for each grade. Composite measures were calculated

Table 1

Reaction times (in ms) and error rates (in %) averaged across items and participants

Reading age level	Suffixed word	Suffixed nonword	Non-suffixed nonword	Unrelated	Number of children
<i>Reaction times (ms)</i>					
Grade 1	1451 (405)	1748 (477)	1721 (371)	1576 (505)	4
Grade 2	1390 (377)	1412 (404)	1410 (359)	1400 (377)	45
Grade 3	1086 (255)	1109 (283)	1112 (279)	1166 (306)	45
Grade 4	961 (184)	971 (159)	967 (158)	982 (178)	31
Grade 5	941 (178)	925 (168)	935 (161)	968 (190)	35
Grade 6	933 (167)	977 (187)	952 (177)	957 (143)	21
Grade 7	808 (158)	801 (141)	860 (211)	887 (117)	8
Grade 8	674 (161)	681 (97)	685 (98)	716 (123)	2
<i>Error rates (%)</i>					
Grade 1	45.0 (6.0)	40.5 (9.5)	40.0 (10.7)	48.0 (5.1)	4
Grade 2	32.0 (18.0)	37.2 (12.9)	33.6 (17.5)	32.5 (16.3)	45
Grade 3	22.4 (18.9)	24.0 (17.8)	21.2 (18.8)	24.9 (19.0)	45
Grade 4	20.7 (17.3)	20.1 (18.4)	16.8 (15.8)	18.4 (17.8)	31
Grade 5	16.0 (16.8)	18.0 (19.0)	15.2 (18.0)	16.6 (18.6)	35
Grade 6	13.6 (14.7)	14.3 (17.5)	17.3 (17.8)	14.9 (16.5)	21
Grade 7	24.5 (16.0)	10.7 (14.7)	19.5 (16.5)	5.1 (14.3)	8
Grade 8	15.1 (21.3)	18.8 (26.6)	23.4 (33.1)	20.2 (28.6)	2

Note. Standard deviations are presented in parentheses. Reading age level is calculated based on the Alouette reading proficiency test (Lefavrais, 2005).

by adding the standard scores of the subtests. Factor grade was included to explore the development of morphological segmentation across primary school grades. Ending frequency was examined as a continuous variable. Finally, to rule out that any of the observed effects were due to differences in stem frequency (i.e., target frequency), we included this factor as a covariate in the analyses.

A linear mixed-effects model as implemented in the lme4 package (Bates, Maechler, Bolker, & Walker, 2014) in the statistical software R (Version 3.0.3; R Development Core Team, 2008) was fitted with eight fixed-effect factors (prime type: suffixed word, suffixed nonword, non-suffixed nonword, or unrelated word; ending frequency; reading proficiency; spelling proficiency; morphological awareness; grade; stem frequency; and trial order), their interactions, and two random-effect factors (random intercepts for participants and items). The p values were determined using the package lmerTest (Kuznetsova, Brockhoff, & Christensen, 2014). A detailed summary of the linear mixed-effects results is reported in Appendix A. RT analyses revealed a significant priming effect in the suffixed word condition ($t = 4.17$, $p < .001$) and the suffixed nonword condition ($t = 2.28$, $p = .022$) as well as a marginally significant priming effect in the non-suffixed nonword condition ($t = 1.85$, $p = .065$) relative to the unrelated control condition. Priming in the suffixed word condition was significantly larger than priming in the non-suffixed nonword condition ($t = 2.33$, $p = .020$) and marginally significantly larger than priming in the suffixed nonword condition ($t = 1.88$, $p = .060$). Although there was a significant main effect of grade ($t = 8.47$, $p < .001$), grade did not modulate priming (see Fig. 1, left panel).

Ending frequency, stem frequency, spelling proficiency, and morphological awareness did not significantly improve the model's fit and, therefore, were excluded from the analyses ($c^2 = 0.69$, $p = .405$; $c^2 = 0.13$, $p = .716$; $c^2 = 0.04$, $p = .847$; and $c^2 = 0.83$, $p = .362$, respectively). Interestingly, however, the results revealed that although priming in the suffixed word condition (relative to the unrelated control condition) was not modulated by reading proficiency ($t = 1.63$, $p = .102$), reading proficiency *did* modulate priming in the suffixed nonword condition and non-suffixed nonword condition ($t = 3.56$, $p < .001$, and $t = 2.62$, $p = .009$, respectively), showing that there was an inhibitory effect for children with low levels of reading proficiency but a facilitatory effect for children with high levels of reading proficiency (see Fig. 1, right panel).

Error analyses followed the same logic as the RT analyses. We applied a binomial variance assumption to the trial-level binary data using the function *glmer* as part of the R package lme4. There was a significant effect of grade ($z = 5.89$, $p < .001$), showing that errors decreased with increasing grade levels. There was a significant effect of reading proficiency ($z = 5.67$, $p < .001$), showing that errors decreased with increasing proficiency. No other effects were significant.

Discussion

In the current study, we tested a large sample of French children in Grades 2 to 5 of primary education to (a) investigate the development of morpho-orthographic and morpho-semantic processing mechanisms across grades and (b) examine changes in morpho-orthographic and morpho-semantic

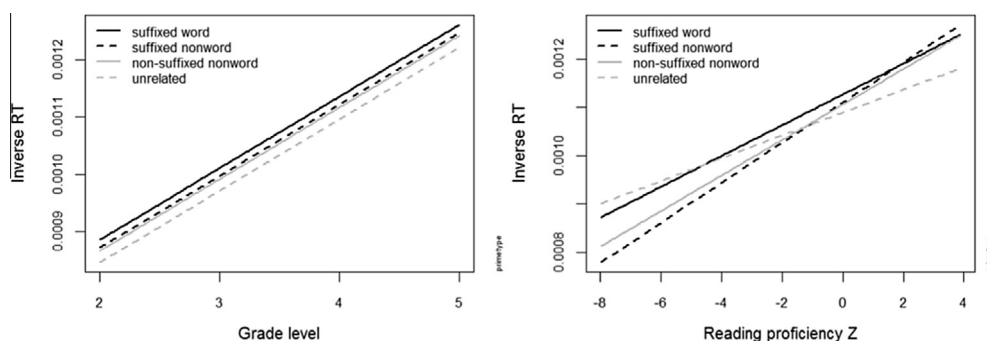


Fig. 1. Inverse reaction times to targets preceded by unrelated, suffixed word, suffixed nonword, and non-suffixed nonword primes as a function of grade (left panel) and reading proficiency (right panel).

processing as a function of individual differences in reading, spelling, and morphological proficiency in beginning readers. Using the masked priming paradigm combined with a lexical decision task, priming in a truly suffixed condition (e.g., *tristesse*–*TRISTE*) was compared with a suffixed nonword condition (e.g., *tristerie*–*TRISTE*) and a non-suffixed nonword condition (e.g., *tristald*–*TRISTE*) using the same targets across conditions. Relative to an unrelated prime condition, robust suffixed word priming was observed across all children independent of grade and proficiency. This suffixed priming effect was significantly larger than the priming effects observed in the suffixed nonword and non-suffixed nonword conditions, suggesting that the semantic interpretability of morphologically complex letter strings helps the segmentation into stem and affix. These results are consistent with previous developmental work from French- and English-speaking primary school children (e.g., Beyersmann et al., 2012; Quémart et al., 2011).

Interestingly, priming in the suffixed and non-suffixed nonword conditions did not change as a function of grade. Moreover, there was no difference between the magnitude of suffixed and non-suffixed nonword priming (see comparable trajectories of both conditions in Fig. 1, left panel). Thus, our data do not provide evidence for morpho-orthographic segmentation but rather speak in favor of children using embedded stem activation mechanisms (Beyersmann, Casalis, et al. (2014)) during early morphological processing (we return to this point in more detail below). Most important, however, our current data suggest that priming did not change as a function of grade, thereby providing no evidence for developmental changes in morpho-orthographic and morpho-semantic processing across primary school, at least not when primes are presented subliminally (cf. Ziegler, Bertrand, Lété, & Grainger, 2014).

Our second key finding is that priming in the suffixed and non-suffixed nonword conditions was modulated by individual differences in reading proficiency (see Fig. 1, right panel). Whereas facilitatory priming effects were observed for high-proficiency children, low-proficiency children showed an inhibitory priming effect. These results suggest that high-proficiency children benefit from the activation of embedded words independent of whether they occur in combination with a real suffix (e.g., *rie*) or a non-suffix (e.g., *ald*). Children with higher levels of reading proficiency would be more expert in mapping sublexical orthography onto whole-word orthographic representations, and this would facilitate the activation of the embedded target word during prime processing (see also Beyersmann, Casalis, et al. (2014), for related evidence from adults). That is, letters in the prime stimulus activate whole-word orthographic representations even when the orthographic overlap between prime and target is only partial (Ziegler, Bertrand, et al., 2014). Higher levels of efficiency in such whole-word processing lead to stronger priming effects from embedded stems independent of whether the stem is accompanied by an affix or not.

The absence of embedded stem priming effects in previous developmental priming studies (e.g., Beyersmann et al., 2012; Quémart et al., 2011) is likely due to the fact that stems were embedded in strings that themselves are real words (e.g., *cashew*–*CASH*). The simultaneous activation of the lexical representations of *cash* and *cashew* generates competition between the two lexical entries and prevents stem–target priming (for similar reasoning, see Beyersmann, Casalis, et al. (2014)).

What might be the cause of the inhibitory priming effect seen with low-proficiency children in the suffixed nonword condition (e.g., *tristerie*–*TRISTE*) and non-suffixed nonword condition (e.g., *tristald*–*TRISTE*)? This result suggests that the nonword primes (*tristerie*/*tristald*) generated some form of competitive process that affected activation of the embedded target word (*triste*). This competition could operate as children attempt to acquire novel lexical representations using orthographic learning (see Ziegler, Perry, & Zorzi, 2014). Successful creation of a new representation for a novel orthographic form (e.g., *tristerie*, *tristald*) would require suppressing the activation of orthographically similar known words (e.g., *tristesse*, *triste*), and this would result in slower responses to the embedded target word following a related nonword prime. These competitive processes would be countered in children with higher levels of reading proficiency by the nonword prime activating the embedded stem more strongly.

It is also important to note that in the current study we found no difference between the suffixed and non-suffixed nonword conditions whatever the grade or reading proficiency of the children. In Beyersmann, Casalis, and colleagues' (2014) study, low-proficiency adults showed priming with suffixed nonwords but not with non-suffixed nonwords. Beyersmann and colleagues argued that morpho-orthographic segmentation processes, such as affix stripping, led to enhanced activation of

embedded stems during the processing of suffixed nonword primes by low-proficiency participants. That is, the drop in efficiency in processing whole-word orthographic information (including embedded words) is compensated by morpho-orthographic segmentation processes. The current results suggest that in primary school children, independent of their reading proficiency, morpho-orthographic processing is not yet sufficiently rapid and/or automatized to affect subliminal (i.e., automatic) priming. In contrast, our results suggest that embedded stems function as lexical representations, not as morpho-orthographic representations, which can be activated subliminally (i.e., automatically) during the process of learning to read. Morpho-orthographic segmentation mechanisms appear to be acquired quite late during reading development and are likely to emerge at later stages throughout middle school or high school. We hypothesize that once morpho-orthographic segmentation has been incorporated into the reading system, it will gradually be used to increasingly compensate for lower proficiency in using lexical whole-word representations, as has been shown to be the case in adults (Beyersmann, Casalis, et al. (2014)).

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Appendix A.

Summary of the statistical output models specifying random and fixed effects. RT = reaction times; RP = reading proficiency; NW = nonword; W = word.

Reaction time data

Formula: inverse RT $\sim$ RP * prime type + trial order + grade + (1|target) + (1|subject)

Prime type reference: unrelated

Random effects:					
Groups	Name	Variance	Std. Dev.		
subject	(Intercept)	5.00E−08	2.24E−04		
target	(Intercept)	6.52E−09	8.07E−05		
residual		8.43E−08	2.90E−04		
Fixed effects:					
	Estimate	Std. Error	df	t-Value	Pr(> t)
(Intercept)	5.87E−04	5.67E−05	2.11E+02	10.34	<.001
prime type[non-suffixed NW]	1.66E−05	8.98E−06	5.96E+03	1.89	.065
prime type[suffixed NW]	2.05E−05	8.99E−06	5.96E+03	2.28	.022
prime type[suffixed W]	3.74E−05	8.98E−06	5.96E+03	4.17	<.001
RP	2.36E−05	9.54E−06	2.32E+02	2.47	.014
trial order	−1.70E−08	9.86E−08	5.97E+03	−0.17	.863
grade	1.26E−04	1.49E−05	1.86E+02	8.47	<.001
prime type[non-suffixed NW]:RP	1.31E−05	5.00E−06	5.96E+03	2.62	.009
prime type[suffixed NW]:RP	1.79E−05	5.02E−06	5.96E+03	3.56	<.001
prime type[suffixed W]:RP	8.22E−06	5.03E−06	5.96E+03	1.63	.102

Prime type reference: suffixed word

<i>Fixed effects:</i>					
	Estimate	Std. Error	df	t-Value	Pr(> t)
(Intercept)	6.24E-04	5.67E-05	2.11E+02	11.00	<.001
prime type[unrelated]	-3.74E-05	8.98E-06	5.96E+03	-4.17	<.001
prime type[non-suffixed NW]	-2.08E-05	8.95E-06	5.96E+03	-2.33	.020
prime type[suffixed NW]	-1.69E-05	8.98E-06	5.96E+03	-1.88	.060
RP	3.18E-05	9.53E-06	2.31E+02	3.34	<.001
trial order	-1.70E-08	9.86E-08	5.97E+03	-0.17	.863
grade	1.26E-04	1.49E-05	1.86E+02	8.47	<.001
prime type[unrelated]:RP	-8.22E-06	5.03E-06	5.96E+03	-1.63	.102
prime type[non-suffixed NW]:RP	4.89E-06	4.97E-06	5.96E+03	0.98	.325
prime type[suffixed NW]:RP	9.65E-06	5.01E-06	5.96E+03	1.93	.054

Prime type reference: suffixed nonword

<i>Fixed effects:</i>					
	Estimate	Std. Error	df	t-Value	Pr(> t)
(Intercept)	6.07E-04	5.67E-05	2.11E+02	10.71	<.001
prime type[suffixed W]	1.69E-05	8.98E-06	5.96E+03	1.88	.060
prime type[unrelated]	-2.05E-05	8.99E-06	5.96E+03	-2.28	.022
prime type[non-suffixed NW]	-3.94E-06	8.98E-06	5.96E+03	-0.44	.661
RP	4.15E-05	9.53E-06	2.31E+02	4.35	<.001
trial order	-1.70E-08	9.86E-08	5.97E+03	-0.17	.863
grade	1.26E-04	1.49E-05	1.86E+02	8.47	<.001
prime type[suffixed W]:RP	-9.65E-06	5.01E-06	5.96E+03	-1.93	.054
prime type[unrelated]:RP	-1.79E-05	5.02E-06	5.96E+03	-3.56	<.001
prime type[non-suffixed NW]:RP	-4.76E-06	4.98E-06	5.96E+03	-0.96	.338

Prime type reference: non-suffixed nonword

<i>Fixed effects:</i>					
	Estimate	Std. Error	df	t-Value	Pr(> t)
(Intercept)	6.03E-04	5.67E-05	2.11E+02	10.64	<.001
prime type[suffixed NW]	3.94E-06	8.98E-06	5.96E+03	0.44	.661
prime type[suffixed W]	2.08E-05	8.95E-06	5.96E+03	2.33	.020
prime type[unrelated]	-1.66E-05	8.98E-06	5.96E+03	-1.85	.065
RP	3.67E-05	9.51E-06	2.29E+02	3.86	<.001
trial order	-1.70E-08	9.86E-08	5.97E+03	-0.17	.863
grade	1.26E-04	1.49E-05	1.86E+02	8.47	<.001
prime type[suffixed NW]:RP	4.76E-06	4.98E-06	5.96E+03	0.96	.338
prime type[suffixed W]:RP	-4.89E-06	4.97E-06	5.96E+03	-0.98	.325
prime type[unrelated]:RP	-1.31E-05	5.00E-06	5.96E+03	-2.62	.009

Error data

Formula: error rates $\sim$ grade + RP + (1|target) + (1|subject)

Random effects:				
Groups	Name	Variance	Std. Dev.	
subject	(Intercept)	0.5834	0.7638	
target	(Intercept)	0.3164	0.5625	
Fixed effects:				
	Estimate	Std. Error	z-Value	Pr(> z)
(Intercept)	−1.10434	0.2315	−4.77	<.001
grade	−0.39539	0.06029	−6.59	<.001
RP	−0.22748	0.03587	−6.34	<.001

Note. RT, reaction time; RP, reading proficiency; NW, nonword; W, word.

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